CMAI 480

Computer Vision

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Reading Assignment

- Szeliski, Chapter 2 (Image Formation)

Primitive Introduction

CHAPTER 2: IMAGE FORMATION

Outline

- Geometric primitives and transformations
  - Review of Linear Algebra
  - Projection geometry
    - Homogeneous Representation of 2D/3D entities or features
    - 2D/3D Image Transformations
  - Camera geometry
- Human vision system
- Human vision for computer vision
- Cameras and image formation

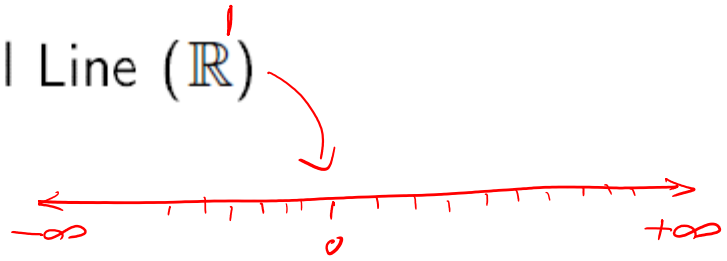
Review Linear Algebra: Outline

- Review: Basics
- Subspace
  - Definition
  - Examples; Null Subspace
- Span
- Basis
- Eigenspace or Eigen Value Decomposition
- Singular Value Decomposition (SVD)
- Outer Product
- Inner Product

Review Linear Algebra

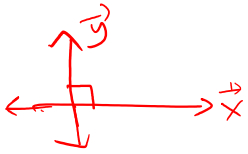
Mathematical Notations: Scalars, Vectors and Matrices

- Recall the Definition of real numbers or real Line ($\mathbb{R}$)



- Recall the Definition of a
  - Scalar
  - Vector (Column vector and row vector)
  - Matrix

- Definition of $\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, where\ x_i \in \mathbb{R}\ \forall\ 1 \leq i \leq n \right\}$
  - What is $\mathbb{R}^1$: line
  - What is $\mathbb{R}^2$: Cartesian Plane



  - What is, $\mathbb{R}^3$: 3D dimensional space

Review Linear Algebra
SUBSPACE

- Definition: Let **S** be a subset of vectors in $\mathbb{R}^n$, then we say **S** is a **SUBSPACE** of $\mathbb{R}^n$ if and only if **S** statisfies 3 conditions:
  - The zero vector belongs to **S**, i.e. $\vec{0} \in S$.
  - S is closure under scalar Multiplication, i.e. if $c \in \mathbb{R}$ and $\vec{x} \in S$, then $c\vec{x} \in S$.
  - S is closure under addition operation, i.e. if $\vec{x}_1, \vec{x}_2 \in S$, then $\vec{x}_1 + \vec{x}_2 \in S$.
- Example: let $S = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right\}$, then is S a valid subspace?
  - Ans: Yes, by applying the 3 conditions above

Review Linear Algebra

SUBSPACE: Example on finding Null-Subspace of a matrix A

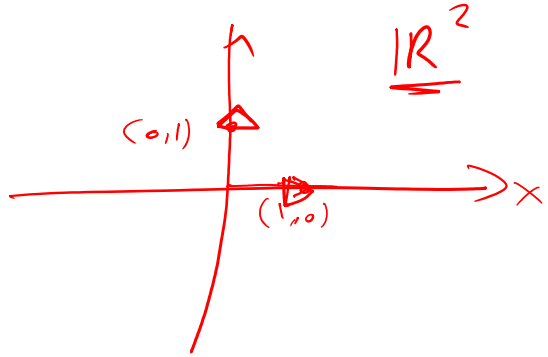
- If we assume that we are given a matrix

$$A_{m\times n} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_n \end{bmatrix}$$, then the following set of vectors to be called **the null space of matrix** A and to be denoted by $N(A) = \{\vec{x} \in \mathbb{R}^n | A_{m\times n}\vec{x}_{n\times 1} = \vec{0}_{m\times 1}\}$ is a valid subspace of $\mathbb{R}^n$.
  - Proof is done by checking the 3 conditions from the definition of the Subspace.
    - $\vec{0} \in N(A)$
    - If $c \in \mathbb{R}$ and $\vec{x} \in N(A)$, then $c\vec{x} \in N(A)$
    - if $\vec{x}_1, \vec{x}_2 \in N(A)$, i.e. $A\vec{x}_1 = \vec{0}$ and $A\vec{x}_2 = \vec{0}$ then $A(\vec{x}_1 + \vec{x}_2) = A\vec{x}_1 + A\vec{x}_2 \in N(A)$

Review Linear Algebra
SPAN

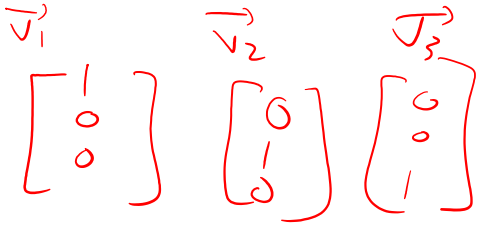
- Definition: we say **SPAN**$\{ \vec{v}_1, \vec{v}_2, \cdots \vec{v}_n \}$ to denote the set of all vectors that can be obtained via a linear combination (e.g. addition and scalar multiplication) from the set of vectors $\{ \vec{v}_1, \vec{v}_2, \cdots \vec{v}_n \}$.
  - i.e. $c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n$, where $c_i \in \mathbb{R} \; \forall \; 1 \leq i \leq n$
- Question: Does the set of the following 2 dimensional vectors $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ span $\mathbb{R}^2$ (i.e. the Cartesian plane)?
  - Anser: Yes

Basis of



Review Linear Algebra

BASIS



- Definition: we say **BASIS** of a given subspace of $\mathbb{R}^n$ to denote the **minimum number of linearly independent vectors** of a given matrix $A_{m\times n}$ that **spans** $\mathbb{R}^n$.
- But, what do we mean by a linearly independent set of vectors for a given matrix A.
  - Let $A_{m\times n} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} = [\ \vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n\ ]$, where the dimension of each vector is $m \times 1$
  - Recall the null space of matrix A: $N(A) = \{\vec{x} \in \mathbb{R}^n | A_{m\times n}\vec{x}_{n\times 1} = \vec{0}_{m\times 1}\}$
  - Now, the set of vectors $\{\ \vec{v}_1, \quad \vec{v}_2, \quad \cdots \quad \vec{v}_n\ \}$ is to be said linearly independent if and only if (*iff*) the only solution $\vec{x} \in \mathbb{R}^n$ to the null space of A is the null or zero vector.

    i.e. if $\vec{x} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$, then $A\vec{x} = [\ \vec{v}_1, \quad \vec{v}_2, \quad \cdots \quad \vec{v}_n\ ] \begin{bmatrix} c_1 \\ \vdots \\ c_2 \end{bmatrix} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n = \vec{0}$, only when $c_1 = c_2 = \cdots = c_n = 0$.

Review Linear Algebra

BASIS: Example

- Question: If we let the matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then what are the basis vectors of A?
  - Solution: let's find the linearly independent vectors or columns of A that span $\mathbb{R}^2$.
  - As such let's specify the null space of A: $N(A) = \{\vec{x} \in \mathbb{R}^2 | A_{2\times 2}\vec{x}_{2\times 1} = \vec{0}_{2\times 1}\}$
  - Let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, then $N(A) = \left\{\vec{x} \in \mathbb{R}^2 | \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}\right\}$. In other words, this only occurs when $x_1 = 0$ and $x_2 = 0$
  - Therefore and indeed $\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$ form the basis vectors of matrix A that span $\mathbb{R}^2$.
    - Note, we have seen that the set vectors $\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$ indeed span $\mathbb{R}^2$

Review Linear Algebra
Payoffs of the BASIS

- The number of basis vectors for a given matrix $A_{m \times n}$ denotes the **RANK** of the matrix.
  - When the number of basis vectors of A equals n, then the matrix is said to be of **Full Rank** or **non-singular** or **invertible.**
    - In such a case the determinant of A must not be equal to zero. i.e. $det(A) \neq 0$

Review Linear Algebra
Rank of a Matrix

- The rank of a matrix A is:
  - The size of the largest collection of linearly independent columns of A (the column rank) or
    - The column rank of A is the dimension of the column space of A
  - The size of the largest collection of linearly independent rows of A (the row rank).
    - The row rank of A is the dimension of the row space of A
- For every matrix, the column rank is equal to the row rank.
- Examples:
  - The matrix $\begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix}$ has rank 2:
    - The first two rows are linearly independent, so the rank is at least 2, but all three rows are linearly dependent (the first is equal to the sum of the second and third) so the rank must be less than 3 (3 is considered as the full rank)
    - When the matrix has full rank, it is said to be non-singular **otherwise** it is said to be singular.

Review Linear Algebra

Rank of a Matrix

$A_{2\times 100}$, $A^T_{100\times 2}$

- A common approach to finding the rank of a matrix is to reduce it to a simpler form, generally row echelon form, by elementary row operations.
  - Row operations do not change the row space (hence do not change the row rank), and also do not change the column rank.
  - Once in row echelon form, the rank is clearly the same for both row rank and column rank, and equals **the number of pivots** (or basic columns) and also the number of non-zero rows.
  - Review Gaussian Elimination (For solving the system $AX = B$)
  - $\text{rank}(A^TA) = \text{rank}(AA^T) = \text{rank}(A) = \text{rank}(A^T)$ $= \text{rank}(\text{rref}(A))$

$B_{2\times 2} = AA^T$

Review Linear Algebra

- Let's now get back to how we can find the null space of an arbitrary matrix $A_{m \times n}$.

Review Linear Algebra

How to find the Null space of a given matrix (1)

- Example: Let $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix}$, find $N(A)$?
  - Solution: $N(A) = \{\vec{x} \in \mathbb{R}^4 \mid A_{3\times 4}\vec{x}_{4\times 1} = \vec{0}_{3\times 1}\}$
  - $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$
  - Then we need to solve the above system or the following 3 equations:
    - $x_1 + x_2 + x_3 + x_4 = 0$
    - $x_1 + 2x_2 + 3x_3 + 4x_4 = 0$
    - $4x_1 + 3x_2 + 3x_3 + x_4 = 0$

Review Linear Algebra

How to find the Null space of a given matrix (2)

- We can alternatively form the augmented matrix and then use Gaussian Elimination in a reduced row echhelon form to a compute a solution:
- The Augmented Matrix: $\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & 4 & 0 \\ 4 & 3 & 2 & 1 & 0 \end{array}\right]$

free variable.

x_1 x_2 x_3 x_4

  - Reduced row echelon form : $rref(A) = \left[\begin{array}{cccc|c} 1 & 0 & -1 & -2 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$
  - Then we have two equations:
  - $x_1 - x_3 - 2x_4 = 0$, which can be written as $x_1 = x_3 + 2x_4$
  - $x_2 + 2x_3 + 3x_4 = 0$, which can also be written as $x_2 = -2x_3 - 3x_4$

- $\begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & 4 & 0 \\ 4 & 3 & 2 & 1 & 0 \end{bmatrix}, \begin{matrix} R_2 = R_2 - R_1 \\ R_3 = R_3 - 4R_1 \end{matrix}$
- $\begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & -1 & -2 & -3 & 0 \end{bmatrix}, R_3 = R_3 + R_2$
- $\begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, R_1 = R_1 - R_2$
- $\begin{bmatrix} 1 & 0 & -1 & -2 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 & -2 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

pivot: x_1 x_2 — free variable: x_3 x_4

rref(A)

$x_3, x_4 \in \mathbb{R}$ real line.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$

$x_1 = x_3 + 2x_4$

$x_2 = -2x_3 - 3x_4$

$x_3 = x_3$

$x_4 = x_4$

$\rightarrow x_1 - x_3 - 2x_4 = 0$

$\rightarrow x_2 + 2x_3 + 3x_4 = 0$

Review Linear Algebra

How to find the Null space of a given matrix (3)

- Based on the last two equations in the previous slide, the solution $\vec{x}$ can be arranged as a function of what to be called the free variables (x_3, x_4) as:
- $$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$$
- x_3 and x_4 are called free variables as they are allowed to take any values from the real number i.e. $x_3, x_4 \in \mathbb{R}$ without affecting our desired solution
- In other words, $N(A) = SPAN\left\{\begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}\right\}$ and $\left\{\begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}\right\}$ are to be called the basis vectors of the null space of the matrix A.
- We should also note that the basis of the matrix A (Known as the basis of the column space of A) are the vectors $\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}\right\}$.

Review Linear Algebra

Important Notes

$$A\vec{x} = \vec{b}$$

$$\vec{x} = A^{-1}\vec{b}$$

only when A is non-singular.

- The number of vectors (not including the null vector) in the null space basis is to be called **the RANK Deficiency** or the dimension of the null space.
- Note that: $N(A) = N(rref(A))$
  - If the null space dimension is greater than zero, then the matrix is to be said **Singular** or **non-invertible** and the determinant of the matrix equals zero i.e. $det(A) = 0$

$N(A) = \{\forall \vec{x} \in \mathbb{R}^n \text{ s.t } A\vec{x} = \vec{0}\}$

Review Linear Algebra

Example on Computing the Rank of a Matrix

- For example, the matrix A given by $A = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix}$ can be put in reduced row-echelon form by using the following elementary row operations:
  - $\begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix} R_2 \to 2R_1 + R_2$
  - $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 3 & 5 & 0 \end{bmatrix} R_3 \to -3R_1 + R_3$
  - $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & -1 & -3 \end{bmatrix} R_3 \to R_2 + R_3$
  - $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} R_1 \to -2R_2 + R_1 \Rightarrow \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$ (x_3 dependent or Free variable; rref(A))

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 = 5x_3$$
$$x_2 = -3x_3$$
$$x_3 = x_3$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}$$

Basis of $N(A)$

- The final matrix (in reduced row echelon form) has two non-zero rows and thus the rank of matrix A is 2.

Review Linear Algebra

SUBSPACE: Another Example – Finding the EIGEN SUBSPACE

- Let's now look into an important subspace that is called the **Eigenspace**.
- The Eigenspace is a valid Subspace of $\mathbb{R}^n$ that is only defined for **non-singular** square matrices A as

$$Eig(A) = \{\vec{v} \in \mathbb{R}^n | A_{n\times n}\vec{v}_{n\times 1} = \lambda \vec{v}_{n\times 1}, \lambda \in \mathbb{R}\}$$

  - $\vec{v}$ is known as an eigenvector
  - λ is known as an eigenvalue corresponding to the eigenvector $\vec{v}$.
    - $\lambda \in \mathbb{R}$
    - λ has many affects on its corresponding eigenvector $\vec{v}$ such as: Elongation ($\lambda > 1$), shortening ($0 < \lambda < 1$), or flipping when λ has a negative sign or value.
    - Every eigenvector has a corresponding and unique eigenvalue.

Review Linear Algebra

EIGENSPACE

- Note that: Number of the Eigenvectors = Rank of the Matrix = number of the basis or independent column vectors of the given matrix.
- As such when we want to find the Eigenspace (known also as eigen-value decomposition) of a given non-singular matrix A, then we meant to find all of it the:
  - Eigenvectors
  - Eigenvalues

Review Linear Algebra
Payoffs EIGENSPACE

- The Eigenspace has many interesting applications in computer vision such as Face Recognition
  - A bounding box is to be drawn around a given face in a given image and then being cropped.
  - The cropped face is to be stored as a matrix of a given size and eigen-value decomposition is being computed. Then the Eigenspace of that face is to be stored along with a given label or tag in a database for face recognition.
  - Note that: Each face should correspond to a unique Eigenspace provided enough dimensionality of representation (This surely will affect the accuracy of the recognition).

Review Linear Algebra
How to compute EIGENSPACE

- How to compute the Eigenspace for a given matrix.
  - Make sure that the given matrix is non-singular
  - Solve the system $\{\vec{v} \in \mathbb{R}^n | A_{n\times n}\vec{v}_{n\times 1} = \lambda \vec{v}_{n\times 1}, \lambda \in \mathbb{R}\}$
    - In other words, we need to solve $A\vec{v} = \lambda\vec{v}$, which can also be written as $A\vec{v} - \lambda\vec{v} = \vec{0}$ or as $(A - \lambda I)\vec{v} = 0$, where I is an identity matrix.
    - That means, recalling the null-space of a given matrix, the set of desired eigenvectors is just the null space of the **singular** matrix $(A - \lambda I)$.
    - But, how we shall compute the set of eigenvalues λ. This will be done using the fact that the matrix $(A - \lambda I)$ should be singular (otherwise we should have an empty set of eigenvectors) and thus the $det(A - \lambda I) = 0$ (This equation is known as the characteristic polynomial of A).
    - Let's see an example.

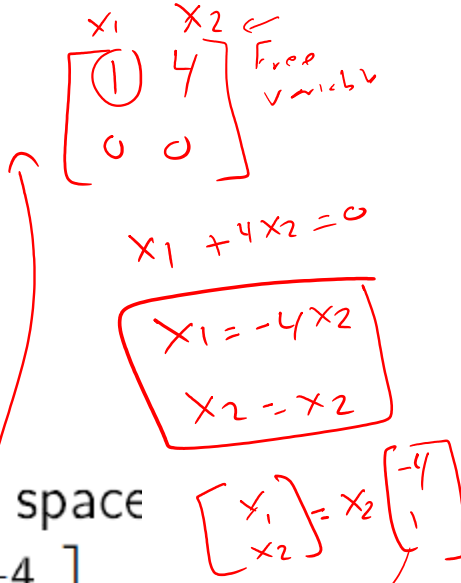
Review Linear Algebra

How to compute EIGENSPACE: Example (1)

- Suppose we have a non-singular matrix $A = \begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix}$, then to find the eigenspace of A:
  - First solve the characteristic polynomial of A. i.e. $det(A - \lambda I) = 0$
    - $det\left(\begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0$
    - $det\left(\begin{bmatrix} 2-\lambda & -4 \\ -1 & -1-\lambda \end{bmatrix}\right) = \begin{vmatrix} 2-\lambda & -4 \\ -1 & -1-\lambda \end{vmatrix} = 0$
    - $(2-\lambda)(-1-\lambda) - (-4)(-1) = 0$
    - $\lambda^2 - \lambda - 6 = 0$
    - $(\lambda - 3)(\lambda + 2) = 0$
    - i.e. Either $\lambda = 3$ or $\lambda = -2$

Review Linear Algebra

How to compute EIGENSPACE: Example (2)



- Now to find the set of eigenvectors, then we need to consider each eigenvalue separately, find the matrix $A - \lambda I$, and compute the null space
  - For $\lambda = 3$, $A - \lambda I = \begin{bmatrix} 2-\lambda & -4 \\ -1 & -1-\lambda \end{bmatrix} = \begin{bmatrix} 2-3 & -4 \\ -1 & -1-3 \end{bmatrix} = \begin{bmatrix} -1 & -4 \\ -1 & -4 \end{bmatrix}$.
    - The null space for the above matrix is computed to be a span of $\left\{ \begin{bmatrix} -4 \\ 1 \end{bmatrix} \right\}$, which is considered as the first eigenvector.
  - For $\lambda = -2$,
    $A - \lambda I = \begin{bmatrix} 2-\lambda & -4 \\ -1 & -1-\lambda \end{bmatrix} = \begin{bmatrix} 2-(-2) & -4 \\ -1 & -1-(-2) \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ -1 & 1 \end{bmatrix}$.
    - The null space for the above matrix is computed to be a span of $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$, which is considered as the second eigenvector.

Review Linear Algebra

EIGENSPACE: Note

- Now, what **if the given matrix A is non-square or simply singular**. Can we still compute the eigenspace of A for a given λ. The answer is yes, but then this will be known as computing the **singular value decomposition** of A (or **SVD(A)**).

Review Linear Algebra

Matrix → orthogonal / orthonormal

SVD

- **Theorem:** Any $n \times m$ matrix A can be written in the form $A_{n\times m} = U_{n\times n}\Sigma_{n\times m}V^T_{m\times m}$ where:
  - U is an orthogonal $n \times n$ matrix,
    - But what is orthogonal and how it is different from being orthonormal (identity is 1)
    - This matrix has special effect to be known as a rotational effect.
  - V is an orthogonal $m \times m$ matrix,
    - This matrix also has has a rotational effect.
  - and Σ is an $n \times m$ matrix whose first r diagonal entries are the non-zero singular values $\sigma_1, \sigma_2, \ldots, \sigma_r$ of A and whose other entries are all zeros.
    - This matrix has a special effect to be known as differential effects (elongation of one dimension and compression on other one).
  - The expression $U\Sigma V^T$ is known as the Singular Value Decomposition of A .

- We will review SVD Further when needed.

Review Linear Algebra
Outer Product

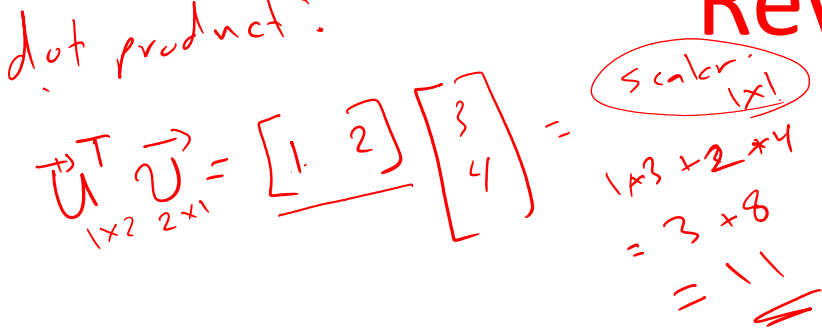
- The outer product $u \otimes v$ is equivalent to a matrix multiplication uv^T.
  - Given the vectors
    - $\mathrm{u} = (u_1, u_2, \ldots, u_m), \mathrm{v} = (v_1, v_2, \ldots, v_n)$
    - u is represented as a $m \times 1$ column vector and v as a $n \times 1$ column vector (which makes v^T a row vector).
  - Their outer product $u \otimes v$ is defined as the $m \times n$ matrix A obtained by multiplying each element of u by each element of v
    - $$\mathrm{u} \otimes \mathrm{v} = \mathrm{A} = \begin{bmatrix} u_1 v_1 & u_1 v_2 & \ldots & u_1 v_n \\ u_2 v_1 & u_2 v_2 & \ldots & u_2 v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_m v_1 & u_m v_2 & \ldots & u_m v_n \end{bmatrix}.$$
- For instance, if $m = 4$ and $n = 3$, then
  - $$\mathrm{u} \otimes \mathrm{v} = \mathrm{uv}^{\mathrm{T}} = \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}}_{4\times 1} \underbrace{\begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}}_{1\times 3} = \underbrace{\begin{bmatrix} u_1 v_1 & u_1 v_2 & u_1 v_3 \\ u_2 v_1 & u_2 v_2 & u_2 v_3 \\ u_3 v_1 & u_3 v_2 & u_3 v_3 \\ u_4 v_1 & u_4 v_2 & u_4 v_3 \end{bmatrix}}_{4\times 3}.$$

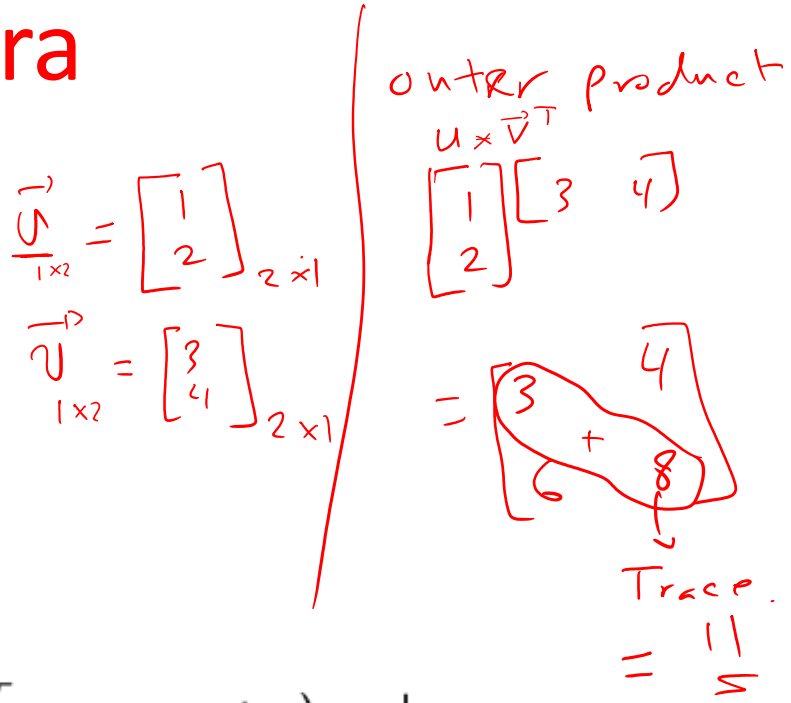
Review Linear Algebra
Outer Product

- Rank of an outer product
  - If u and v are both nonzero then the **outer product matrix** uv^T **always has rank 1**.

Review Linear Algebra

Inner Product

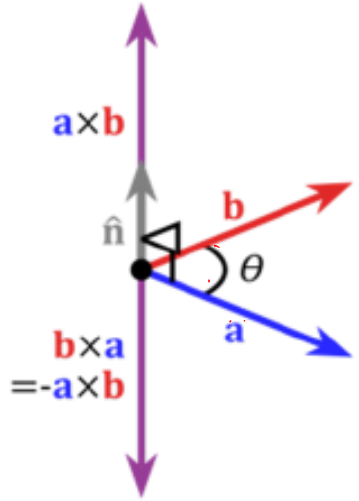




- Inner product (Better known as the dot product)
- Given the vectors
  - $u = (u_1, u_2, \ldots, u_m), v = (v_1, v_2, \ldots, v_n)$
  - u is represented as a $m \times 1$ column vector (which makes u^T a row vector) and v as a $n \times 1$ column vector.
  - If $m = n$, then one can take the matrix product the other way, yielding a scalar (or 1×1 matrix): $\langle u, v \rangle = u^T v = \|\vec{u}\| * \|\vec{v}\| \cos\theta$
  - The inner product is the **trace** (sum of the elements on the main diagonal; the diagonal from the upper left to the lower right) of the outer product matrix.

Review Linear Algebra

Dot Product Vs. Cross Product



$$\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta,$$

$$\mathbf{a}\times\mathbf{b} = \det\begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix} = \mathbf{i}(a_2 b_3 - a_3 b_2) - \mathbf{j}(a_1 b_3 - a_3 b_1) + \mathbf{k}(a_1 b_2 - a_2 b_1) = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ -(a_1 b_3 - a_3 b_1) \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

Cross Product

- We note that with $\vec{A} = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix}$ and $\vec{B} = \begin{pmatrix} b_x \\ b_y \\ b_z \end{pmatrix}$, the vector cross product $\vec{A} \times \vec{B}$ can be expressed as the matrix-vector product $[A]_\times B$, where: $[A]_\times = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}$.
  - Note: $[A]_\times$ is a skew symmetric matrix (det=0)

Computing Cross Product in Matlab

```
> x = [1; 2; 3]
> y = [4; 5; 6]
> z = cross(x, y)
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END OF THE LINEAR ALGEBRA REVIEW